



095946 ADVANCED ALGORITHMS AND PARALLEL PROGRAMMING

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a.a. 2026-2027

COURSE OBJECTIVES

- This course deals with advanced topics in algorithm design and parallel programming.
- The course is structured in two parts. The first part focuses on general methods and algorithms that are not usually covered by the course “Algoritmi e Principi dell’Informatica”, such as, randomization, amortization, approximation algorithms, string searching/matching, etc.
- The second part deals with parallel programming: Automatic vs. Manual Parallelization, Parallelizing Compilers, Parallel Patterns, Partitioning (domain vs functional decomposition), Communication (cost, latency, bandwidth, visibility, synchronization, etc.), Data Dependencies and Tools/languages such as OpenMP and MPI (no more on **CUDA**).

COURSE LEARNING OUTCOME: LIFELONG LEARNING SKILLS

- Students will understand how a complex algorithm, possibly parallel, have to be analyzed, designed and assessed. They will play with real problems understanding where pitfall may come when you move from a theoretic formulation down to a real implementation taking into account existing tools and real architectures.

PART I-A - GENERAL METHODS AND ALGORITHMS

Course Objectives and Introduction

Randomized algorithms:
Las Vegas and Monte Carlo algorithms.
Analyzing Randomized algorithms.

Hiring Problem and
Generating Random Permutations

Randomized Quicksort.
Worst-case analysis.
Average-case analysis

Order Statistics,
Randomized divide-and-conquer algorithm

Primality test. Fast exponentiation. Secret key and cryptosystems

PART I-B - GENERAL METHODS AND ALGORITHMS

Karger's Min-Cut Algorithm. Faster version by Karger and Stein

Randomized data structures: Skip Lists, Treaps

Dynamic Programming: Memoization.
Examples of Dynamic Programming: String Matching, BDDs, etc.

Amortized Analysis: Dynamic tables, Aggregate method, Accounting method and Potential method.

Approximate programming

Competitive Analysis
Self-organizing lists
Move-to-front heuristic

PART II - ON PARALLELIZATION

- Design of Parallel Algorithms - Parallel Algorithms and Parallel Programming.
- Parallel Machine Model
- Introducing parallel patterns: Reduce, Split, Compact / Expand and Parallel Prefix Sum.
- More on parallel patterns: Segmented Scan, Sort, Map-reduce, Kernel Fusion.
- Tools and languages for parallel programming: Posix Threads, OpenMP, Message Passing Interface.
- Comparison of Parallel Programming Technologies.
- Optimizing parallel performance.
- Halide: high-performance image and array processing code.

PREVIOUS KNOWLEDGE



Operations
Research



Basic data
structures



Computational
Complexity theory



C/C++ knowledge
could help

AAPP TEACHING MATERIAL

- Introduction to Algorithms, T. Cormen, C. Leiserson, R. Rivest and C. Stein, MIT Press, Cambridge; 3rd edition (20 Aug 2009) ISBN-10: 0262533057, ISBN-13: 978-0262533058
- Randomized Algorithms, Rajeev Motwani, Prabhakar Raghavan. Cambridge University Press (August 25, 1995) ISBN-10: 0521474655, ISBN-13: 978-0521474658
- Michael McCool, James Reinders, Arch Robison, Structured Parallel Programming: Patterns for Efficient Computation, Editore: Morgan Kaufmann, Anno edizione: 2012, ISBN: 0124159931 <http://parallelbook.com/>
- An Introduction to Parallel Programming, Peter Pacheco, Publisher: Morgan Kaufmann; 1 edition (January 21, 2011), ISBN-10: 0123742609, ISBN-13: 978-0123742605
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- Parallel programming online material [link](#)
- Additional material is available on the WeBeep platform of Politecnico di Milano (access restricted to course participants): <https://webeep.polimi.it>

EXAM HOW

- Evaluation is based on a written exam
- Exam content
 - The solution of some problems based on the practical application of the course concepts and techniques
 - Open answers to some questions on the course concepts and techniques
- After each written test, the teacher can complement the assessment procedure with an oral examination



CONTINUOUS ASSESSMENT

- Continuous assessment will be implemented through two intermediate tests: one as a **mid-term** test and one at the **end** of the semester: **Nov. 5 – Dec 17**.
- All students are admitted to the second test, regardless of the outcome of the first one
- The achieved results will be valid till the end of the academic year or till a student ask to repeat the given part
- Each intermediate test contributed to the final grade with **16** points
- The exam is considered passed if, in both parts, the students get a grade not less than **7**, and the sum of the two grades is greater or equal to **18**
- **30 cum laude** is assigned if students get a sum of grades greater than **30**
- If either the first or the second test has a grade less than **7** or the total is less than **18**, the student has to take the written test on one of the following dates according to the schedule provided by the School's Academic Calendar
- The student may use one of the valid partial results on the next exam dates. In this case, a customized written exam version will be provided to the student for the parts not yet valid

INNOVATIVE-LEARNING CLASSROOM ACTIVITIES

- Participation in innovative-learning classroom activities will be assessed and will contribute to the final evaluation grade. In particular, in case of valid grades, some questions of the written exam could be skipped.

LECTURERS

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